Teaching a student who forgets

A plain-language guide to our poster

1. The point of the poster

Students forget. A student who knew a topic on Monday may not remember it after a week off. But many tutoring programs choose the next question as if the student never forgets: once the student answers a topic well, the program moves on and never looks back.

We built a simulated student who forgets, and tested different ways of choosing questions. We are making two claims:

> **Claim 1.** No single way of tutoring is best for every break length. A setting that works well for a short break works badly for a long one.
>
> **Claim 2.** We can measure how far each way of tutoring is from the best possible, at each break length.

Why this matters. The goal of our thesis is a new way of choosing questions that plans for forgetting. These two claims show why such a method is needed, and how much there is to gain.

What "best possible" means here. We tried five tutoring settings. At each break length, the best possible is the score of whichever setting did best at that length. It is a bar we can really reach, not a perfect score.

2. How we tested it

The simulated student

Real students cannot be tested again and again with the same starting point. A simulated student can. Every tutor gets exactly the same simulated students, so the comparison is fair.

- For each topic, the student has a hidden "how well I know this" number from 0 (not at all) to 1 (perfectly).
- A right answer raises it a lot. A wrong answer raises it a little. Harder questions are harder to get right.
- A student learns a topic more slowly if the topics it depends on are weak.
- The student sometimes guesses right (25% of the time) and sometimes slips on something they know.
- **Forgetting:** with no practice, the number slowly falls. The *half-life* is how long it takes to lose half. For our average student it is 14 days. We also tried students with 7 days (forgetful) and 28 days (good memory).
- The tutor never sees the hidden number. It only sees answers, and keeps its own guess about what the student knows.

The practice material (the “dataset”)

We did not use real student records. The material is made up: **40 topics** (named A01, B02 and so on) with **290 questions** in **7 levels**. Some topics must be learned before others. The topic names mean nothing, so the results do not depend on any school subject.

One test

- The student practises in 20 daily sessions of 25 questions (500 questions in all).
- Then comes a **break** of 0 to 28 days with no practice.
- We count the **usable topics**: topics the student still knows at least 50%. The most possible is 40.
- We used 50 simulated students per case, never used while we were designing the tutors. We trust a difference only if it shows up across most students, not just on average.

The tutors we compared

- **CurriculumTutor as published** (the method from the paper we are improving). Its guess about the student never fades. Once it thinks a topic is learned, it stays learned.
- **Fading guess.** The same tutor, but its guess about each topic slowly falls as time passes, as the student's memory does. “Move on at 0.7 / 0.8 / 0.9” means how sure the tutor must be before it stops teaching a topic. A higher number means more practice per topic, and so fewer topics covered, but a safer margin.
- **Adaptive review.** Goes back to topics whose guess has fallen.

3. What we found

Finding 1: a tutor that plans for forgetting does much better

After a 3-day break, the published CurriculumTutor leaves the average student with **22.7** usable topics out of 40. The same tutor with a fading guess leaves **37.0**.

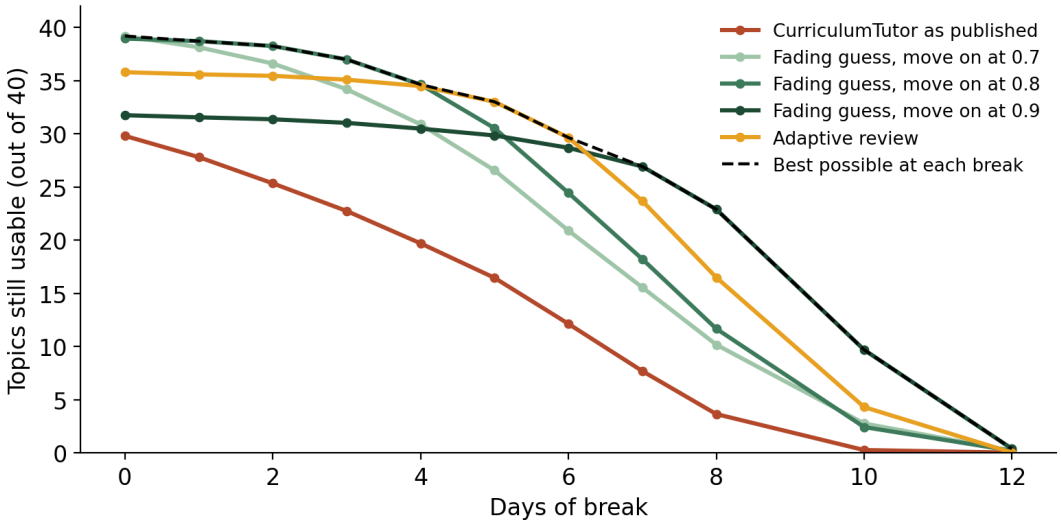
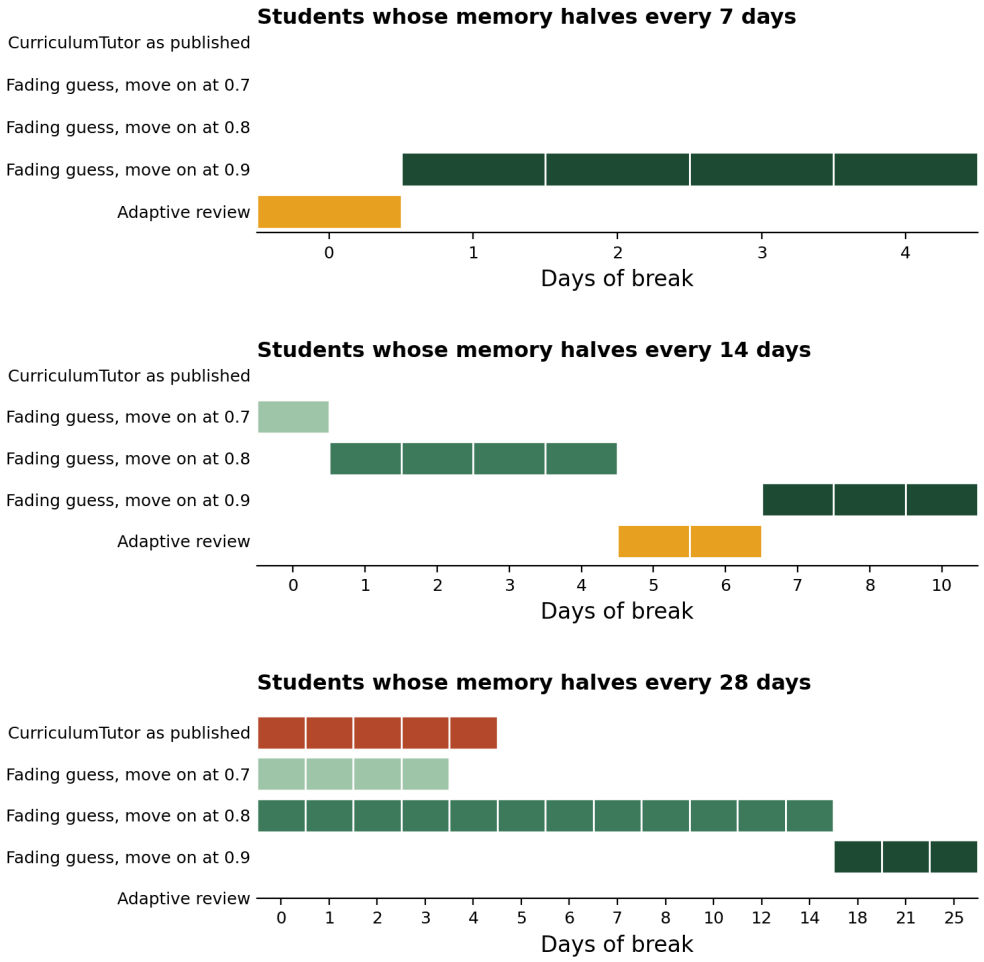


Figure 1. Average student (memory half-life 14 days). Higher is better. The dashed black line is the best possible at each break length.

Finding 2: the best setting changes with the break length (Claim 1)

Look at Figure 1. No coloured line stays on the dashed line the whole way. For the average student, moving on at 0.7 is best with no break, 0.8 is best after 1 to 4 days, adaptive review is best at 5 to 6 days, and moving on at 0.9 is best at 7 to 10 days. A tutor that is careful about every topic loses on short breaks because it covers fewer topics. A relaxed tutor loses on long breaks because the topics fade.



A coloured block means this setting is the best (or tied for best) at that break length.

Figure 2. Which setting is best at each break length, for three kinds of student. The best setting moves to longer breaks as memory gets longer, but it keeps changing in all three.

Finding 3: how much does one fixed choice cost? (Claim 2)

If a tutor must pick one setting and cannot know the break length, how far is it from the best possible? For the average student, even the best single choice (adaptive review) loses **2.64** topics on average and **6.4** at its worst. The published CurriculumTutor loses **14.43** on average and up to **19.24**.

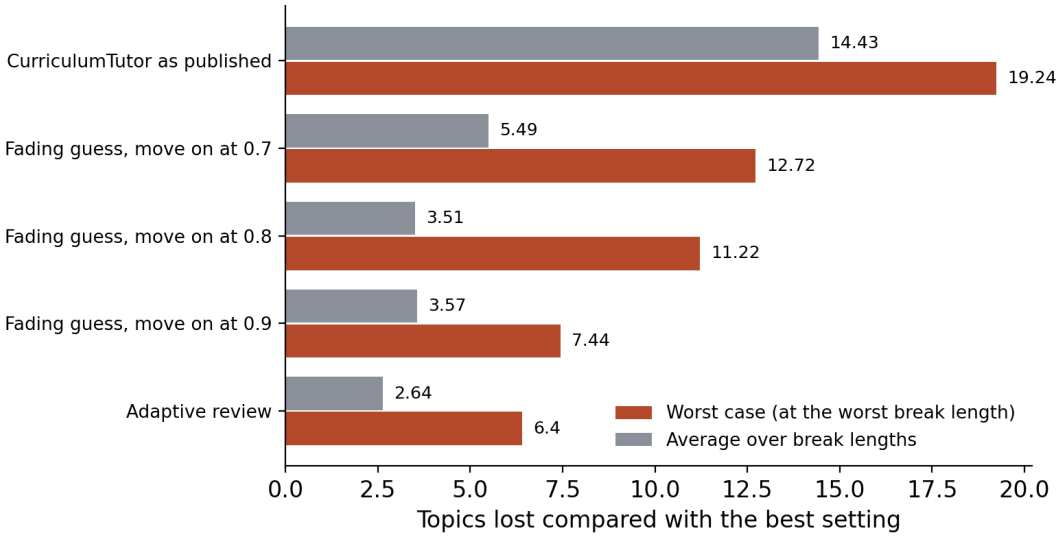


Figure 3. Topics lost by each setting compared with the best setting, average student. Lower is better.

This gap is the room for improvement. A tutor that adapts to the break length could win back part of it.

Finding 4: even a tutor that knows what the student remembers is not the best

We also tried a tutor that is told exactly what the student knows. Across many break lengths it does better (28.2 against 25.6). But after a short 3-day break it does *worse* than the fading guess (35.1 against 37.0). The fading guess is a little cautious, and that caution helps. So "knowing exactly" is not a ceiling, and that is why we measure against the best setting instead.

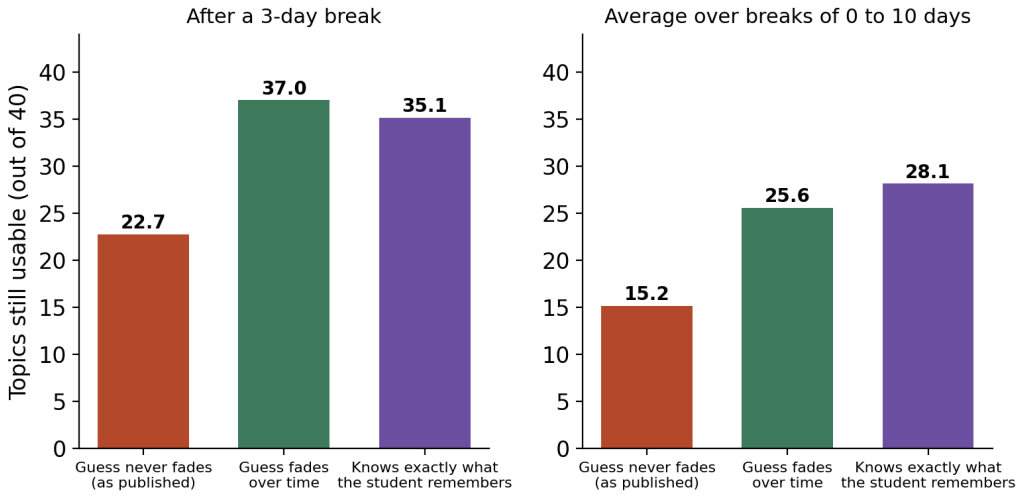


Figure 4. A tutor that knows exactly what the student remembers, compared with the other two.

Finding 5: popular review rules from research did not add anything

We tried three review rules from published work on spaced repetition (review the topic with the lowest recall, review the one closest to a threshold, review in proportion to how much is forgotten). On fresh students they scored within 0.15 topics of the fading-guess tutor, which is

no real difference. So simply adding a review rule is not enough. How much safety margin to keep matters more.

4. What this means, and what we did not show

- Next step of the thesis: a question-picking method that sets its margin and review plan for forgetting, judged over a range of possible breaks, without being told the break length.
- These are **simulated** students and a **made-up** curriculum. They show how tutoring methods behave. They do not prove anything about real students.
- “Best possible” is the best of five settings we tried, picked after seeing the results. A better setting might exist.
- The review rules are our own versions fitted to a curriculum with prerequisites, not the exact methods of the papers.
- The fading guess assumes 5% forgetting per day. For forgetful students (true rate near 9%) more margin helped; this shows that a wrongly assumed forgetting speed has a cost.

Words used

Break: days with no practice. **Usable topic:** a topic the student knows at least 50%. **Half-life:** days for the student's knowledge of an unpractised topic to halve. **Guess (belief):** what the tutor thinks the student knows. **Move on at 0.8:** the tutor stops teaching a topic once its guess reaches 0.8 out of 1.

5. References

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All numbers come from the code and results in this project (results/poster_gap, results/forgetting_rules).